RESEARCH PROPOSAL

How Sensitive Is On-Time Performance to Measurement Choices?

A Robustness Protocol for Public Transit KPIs

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MOTIVATION

SEPTA says a bus is on time if it's within 5 minutes of schedule.
What if that window were 3 minutes? What if they only measured at certain stops?
The number changes each time — but it's reported as a single fact.

1

Opaque numbers

Agencies report OTP as one clean figure, hiding the definitional choices behind it.

2

Incomparable metrics

Two agencies both report "OTP" but measure it differently — comparisons are meaningless.

3

No sensitivity standard

Transit agencies do not appear to use a standard protocol for reporting how sensitive their KPIs are to definition choices.

Same metrics. Different rules.

Publicly documented OTP definitions from four U.S. transit agencies:

Agency	Late Threshold	Early Threshold	Measured At	Cancellations
Spokane Transit	+5 min	**0 min (strict)**	Scheduled timepoints / departure times	Not stated
MTA New York	>5 min late	Any early = OK	Terminal locations	**Counted as late**
Metra (Chicago)	**+6 min**	Not clearly stated	Final stop only	**Excluded**
SEPTA (Philadelphia)	**Not published**	**Not published**	**Not specified**	**Not specified**

SEPTA's public dashboard does not appear to document its OTP methodology. Reported OTP values are not directly comparable across agencies unless methodology is made explicit.

Sources: Metra OTP Report; MTA 2020 Annual Performance Metrics; Swiftly/Spokane Transit webinar; SEPTA Tracking Progress dashboard

A KPI Robustness Protocol

Instead of computing OTP once, compute it many times under documented specification choices and report the full range.

1

Define the Specification Set

Enumerate 6-10 reasonable choices: lateness threshold, stop selection, cancellation handling, stale-data rules

2

Compute KPIs Under All Specs

Run SEPTA GTFS + live GTFS-RT data through every scenario combination

3

Report with Robustness Band

e.g. OTP = 82% ± 4pp (illustrative) — and identify which choices drive the most variance

Research output

Adapting specification-curve analysis (from the reproducibility literature) to transit KPI reporting. The output: (1) a sensitivity analysis of OTP under defensible specifications, (2) identification of the assumptions that matter most, and (3) a proposed minimum reporting standard for transit KPIs.

Current Progress

DATA SOURCES

GTFS Static Schedule

Full SEPTA network downloaded, explored

GTFS-RT Live Feeds

Bus trip updates, vehicle positions, alerts — archiving now because SEPTA has no historical archive

Source: SEPTA (Philadelphia)

Publicly available via Data.gov + versioned GitHub releases

LIVE INFRASTRUCTURE

● LIVE — Archiver running now

```
fetcher-1 | Interval: 60s
[20260228T221026Z] Saved 347,547 bytes
→ /data/raw/gtfs_rt/bus_trip_updates/
[20260228T221126Z] Saved 351,203 bytes
→ /data/raw/gtfs_rt/bus_trip_updates/
[20260228T221226Z] Saved 349,876 bytes
→ /data/raw/gtfs_rt/bus_trip_updates/
```

→ ***Next step: scope 2–3 bus routes, then formally define the specification set***

TIMELINE & RISKS

NOW

Phase 1

Scope + Static Analysis

- → Pick 2–3 routes
- → Define spec set
- → Explore schedules

NEXT

Phase 2

Parse GTFS-RT

- → Decode GFTS-RL Feed Data
- → Match trips
- → Reconstruct events

MID

Phase 3

Compute & Compare

- → OTP under all specs
- → Sensitivity bands
- → Variance decomposition

FINAL

Phase 4

Results + Paper

- → Formulate findings
- → Reporting template
- → Reproducible repo

KEY RISKS & MITIGATIONS

No SEPTA GTFS-RT archive

Self-archiving from week 1; raw protobuf stored for reproducibility

Prediction vs. actual ambiguity

Define two KPI families explicitly (prediction-based vs. observation-based)

Scope creep (too many KPIs)

Start with OTP only; add headway metrics only if time allows

EXPECTED CONTRIBUTION

Research questions

RQ1: How much does reported on-time performance change when we vary reasonable preprocessing and definition choices in SEPTA GTFS / GTFS-RT data?

RQ2: Which specification choices explain the largest share of that variation?

Sensitivity Analysis

Show how much OTP shifts across defensible specification choices, using real SEPTA data

Driver Identification

Determine which preprocessing decisions (threshold, stop selection, cancellations) matter most

Reporting Standard

Propose a minimum transparency protocol for publishing transit KPIs with documented sensitivity

Supporting artifacts: reproducible pipeline (GitHub) · robustness dashboard · one-page protocol checklist

Thank you — happy to take questions.